on the threshold of a vast and awesome Universe that utterly dwarfs —in time, in space, and in potential—the tidy anthropocentric proscenium of our ancestors. We gaze across billions of light-years of space to view the Universe shortly after the Big Bang, and plumb the fine structure of matter. We peer down into the core of our planet, and the blazing interior of our star. We read the genetic language in which is written the diverse skills and propensities of every being on Earth. We uncover hidden chapters in the record of our own origins, and with some anguish better understand our nature and prospects. We invent and refine agriculture, without which almost all of us would starve to death. We create medicines and vaccines that save the lives of billions. We communicate at the speed of light, and whip around the Earth in an hour and a half. We have sent dozens of ships to more than seventy worlds, and four spacecraft to the stars. We are right to rejoice in our accomplishments, to be proud that our species has been able to see so far, and to judge our merit in part by the very science that has so deflated our pretensions.

To our ancestors there was much in Nature to be afraid of—lightning, storms, earthquakes, volcanos, plagues, drought, long winters. Religions arose in part as attempts to propitiate and control, if not much to understand, the disorderly aspect of Nature. The scientific revolution permitted us to glimpse an underlying ordered Universe in which there was a literal harmony of the worlds (Johannes Kepler's phrase). If we understand Nature, there is a prospect of controlling it or at least mitigating the harm it may bring. In this sense, science brought hope.

Most of the great deprovincializing debates were entered into with no thought for their practical implications. Passionate and curious humans wished to understand their actual circumstances, how unique or pedestrian they and their world are, their ultimate origins and destinies, how the Universe works. Surprisingly, some of these debates have yielded the most profound practical benefits. The very method of mathematical reasoning that Isaac Newton introduced to explain the motion of the planets around the Sun has led to most of the technology of our modern world. The Industrial